JAYADEEP GOPINATH

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SUMMARY

MS Computer Science student (Illinois Tech, May 2027) who builds GenAI agents and distributed backends, measured with evaluation harnesses and load tests. Seeking applied AI, forward deployed, and software engineering roles.

EDUCATION

Illinois Institute of Technology · Chicago, IL *Aug 2025 – May 2027*
Master of Science in Computer Science · GPA: 3.50 / 4.0 · KC Chan & Family Scholarship
Relevant coursework: Distributed Systems, Advanced Algorithms, Machine Learning, Advanced Database Organization, Software Engineering

Visvesvaraya Technological University · Bengaluru, India *Dec 2020 – Jul 2024*
Bachelor of Engineering in Computer Science · IJTRE-published research · Government of Karnataka (KSCST) recognition

TECHNICAL SKILLS

GenAI & Agents: LLM APIs (Anthropic Claude, OpenAI), agentic RAG pipelines, tool-calling agent loops, Model Context Protocol (MCP), embeddings, hybrid retrieval (dense + BM25 + RRF), cross-encoder reranking, RAGAS-style evaluation harnesses, PyTorch
AI-Assisted Development: Claude Code, Cursor, GitHub Copilot with prompt-and-verify discipline and regression tests for LLM-generated code
Languages: Python (FastAPI, Flask, pytest, asyncio, SQLAlchemy), Go, TypeScript, JavaScript, Java, C++, C, SQL
Backend & Frontend: Node.js, Express, REST APIs, JWT, OAuth, React, TypeScript, Tailwind CSS, Socket.io, GraphQL (familiar), gRPC (familiar)
Databases: PostgreSQL (pgvector, HNSW indexes), MongoDB (compound indexes), MySQL, Redis, SQL query optimization, schema design
Testing & Observability: pytest, go test, k6 load testing, GitHub Actions CI/CD, Git, Prometheus, Grafana, structured logging
Cloud & Infrastructure: Docker, docker-compose, AWS (EC2, S3, RDS, Lambda basics), Kubernetes basics, Linux

EXPERIENCE

Research Assistant - Illinois Institute of Technology · Chicago, IL *Aug 2026 – Present*
- **Architected agentic RAG pipelines** using Anthropic's Model Context Protocol (MCP), hybrid dense-sparse retrieval with Reciprocal Rank Fusion, and cross-encoder reranking, achieving **p50 end-to-end latency of 293 ms** across multi-step reasoning workflows for research knowledge-base queries
- **Engineered automated evaluation frameworks** with RAGAS-style metrics (faithfulness, answer relevance, context precision, context recall) against a 20-question golden set, enabling continuous quality validation of LLM outputs across evolving research pipelines and improving measurement reliability by 100% over ad-hoc testing
- **Developed deterministic reproducibility tooling** with content-addressable BLAKE2b-seeded embedding fallbacks and pinned model versioning, enabling fully offline CI runs for research experiments without external API dependencies, eliminating network-dependent test flakiness and reducing experiment iteration time by ~40%
- **Built cross-functional research workflows** integrating LLM APIs (Anthropic Claude, OpenAI), PostgreSQL with pgvector HNSW, Python/FastAPI orchestration, and Prometheus/Grafana observability, enabling data-driven methodology decisions across bi-weekly technical reviews

Event Coordinator & Content Creator - Product Management Club, Illinois Institute of Technology · Chicago, IL *Sep 2025 – Feb 2026*
- Prototyped **Python + LLM demo tools** (evaluation harnesses, internal knowledge bots) with technical write-ups for cross-functional audiences; coordinated campus events

Software Engineer Intern - Techfortune · Bengaluru, India *Aug 2023 – Oct 2023*
- Built **Python automation scripts** (pytest, FastAPI) validating REST API responses and ML model outputs across **300+ test records**
- Refactored a legacy Python codebase for readability and testability, cutting average test execution time by **~40%** and reducing CI flakiness
- Designed **SQL schemas with composite indexes and views** for QA dashboards; shipped features end to end in an Agile team of 4 with code reviews

Undergraduate Research Assistant - East West Institute of Technology (CS Dept) · Bengaluru, India *Jan 2021 – Jun 2024*
- Led a multi-year **healthcare AI research project (Naari Shakthi)**: methodology design, prototype, benchmark validation; built a **Python Flask ML microservice** (REST endpoints, automated assertion checks against clinical benchmarks) and a React + Node.js + MongoDB prototype with RBAC
- Co-published IEEE-format paper (IJTRE, Jan 2024); recognized by **Government of Karnataka (KSCST)** for improving rural healthcare access

PROJECTS

Agentic-RAG-MCP · GenAI agent server + LLM evaluation · open-source (MIT) *github.com/JayadeepGopinath/Agentic-RAG-MCP*
Python · FastAPI · Anthropic Claude / OpenAI · Model Context Protocol · PostgreSQL + pgvector · HuggingFace Transformers · Docker · GitHub Actions
- Shipped an **agentic RAG server via Anthropic's Model Context Protocol**, installable in Claude Desktop, Cursor, Cline, Continue, and Zed; multi-step agent loop (plan → search → rerank → cite → decide, up to 5 tool calls); **hybrid retrieval** with cross-encoder reranking
- Built an automated **RAGAS-style evaluation harness** (self-QA) on a 20-question golden set, with an offline mode for CI without secrets or network

Notification-System-Go · Distributed low-latency service in Go · open-source (MIT) *github.com/JayadeepGopinath/Notification-System-Go*
Go · Redis Streams · PostgreSQL · Prometheus · Grafana · Docker · k6 load testing
- Owned the service from design to load test: **170 req/s sustained, p50 17 ms, p95 349 ms at 500 concurrent VUs, 98.77% success rate** across 40,866 requests
- Implemented **at-least-once delivery** (Redis Streams consumer groups), exponential-backoff retries, DLQ, and Postgres-backed idempotency; profiled Postgres pool saturation and Redis XAdd contention and documented the fix path

CodePulse · Full-stack sandboxed code execution platform *github.com/JayadeepGopinath/codepulse*
React · TypeScript · Node.js · Express · Socket.io · Redis · MongoDB · Docker
- Built a full-stack platform for **sandboxed Docker code execution** across 5 languages (CPU/memory limits, JWT-authenticated REST APIs) sustaining **500+ concurrent sessions**; Redis Pub/Sub real-time sync; MongoDB compound indexes for **sub-50 ms** queries

PUBLICATIONS

Jayadeep G - "Smart Sanitary Pad and Medicine Vending Machine with Automated Dispensing, Disposal, and Notification System," *IJTRE Vol. II Issue 5, January 2024, ISSN 2347-4718.*